

AgriSLM Impact Statement

South Africa + Sri Lanka Joint Agricultural Small Language Model TRI AI - Impact Statement Card

Final Draft - aligned with the AgriSLM Values-Led Problem Statement, Research Framework and Ethical Dataset Data Card

1. Purpose of the Impact Assessment

This impact assessment anticipates the broader societal, ethical, environmental and economic implications of AgriSLM. It is intended to guide development toward genuine community benefit while identifying risks that could emerge during research, deployment or future expansion.

2. Potential Positive Impacts

- Improved access to localized and scientifically grounded agricultural information for selected resource-constrained farming communities.
- Greater accessibility through appropriate local languages and offline or low-connectivity deployment.
- Better interpretation of trusted climate and agrometeorological information for crop, soil, water and production decisions.
- Support for sustainable practices relating to soil conservation, water use, climate resilience and responsible crop management.
- Preservation of useful local agricultural knowledge when it is contributed with consent and clearly distinguished from scientifically validated evidence.
- Stronger farmer autonomy by supporting informed decisions while keeping farmers, extension practitioners and agricultural experts central to decision-making.

3. Potential Risks and Unintended Consequences

- Incorrect or outdated agronomic guidance could contribute to crop loss, environmental damage or financial harm.
- Unsafe pesticide, fertilizer or chemical advice could create human, crop or environmental risks.
- Farmers could over-rely on AI and reduce consultation with extension officers or other qualified experts.
- Unequal representation of countries, regions, crops or languages could make the system less reliable for underrepresented communities.
- Traditional or community knowledge could be extracted, decontextualized or used beyond the consent under which it was shared.
- A system designed to reduce the digital divide could unintentionally reinforce it if required hardware, language support or interfaces remain inaccessible.
- Historical and socioeconomic variables such as land tenure could create harmful stereotypes if used as proxies for farmer capability or eligibility.

4. Values Impact

Value	Expected Impact and Responsibility
Fairness	Evaluate performance across represented regions, crops and languages and document exclusions rather than assuming equal performance.
Dignity	Treat farmers as stakeholders and knowledge holders, not merely as data sources or passive recipients.
Autonomy	Provide decision support rather than automated decisions, and communicate uncertainty and limitations.
Community well-being	Prioritize understandable, safe and locally useful agricultural information that can complement existing support systems.
Scientific integrity	Ground high-consequence advice in authoritative sources and distinguish scientific evidence from unvalidated local observation.
Cultural preservation	Respect consent, attribution and context when incorporating local or traditional knowledge.
Sustainability	Favor knowledge that supports responsible soil, water and environmental management.

5. Risk Mitigation and Accountability

- Use authoritative agricultural and climate sources with provenance, update dates and validation status.
- Apply risk levels to agricultural queries; high-risk chemical or pesticide guidance requires authoritative grounding.
- Communicate uncertainty and escalate unsupported or high-consequence questions to appropriate agricultural experts or authorities.
- Keep a held-out evaluation dataset and assess agronomic correctness, groundedness, regional relevance, hallucination, safety and language accessibility.
- Collect farmer or community knowledge only through appropriate consent and governance processes.
- Maintain dataset and knowledge-base versioning, correction and removal procedures.
- Include farmer, extension-practitioner and agricultural-expert feedback in evaluation before broader deployment.
- Clearly state that AgriSLM does not replace extension services, independently forecast weather, guarantee yields, or resolve structural issues such as land, finance and infrastructure.

6. Submission-Ready Impact Statement (300-500 words)

AgriSLM is intended to improve access to localized, scientifically grounded and climate-aware agricultural information for selected resource-constrained farming communities in South Africa and Sri Lanka. Its potential societal benefit lies in reducing agricultural information inequality, particularly where farmers face language, connectivity, extension-access or resource constraints. A compact, offline-capable system could help farmers interpret trusted agricultural and climate information relating to crops, soil, water, pests, diseases, fertilizer and sustainable production. Local-language support and the respectful inclusion of consented local knowledge may also strengthen accessibility, cultural recognition and farmer autonomy.

These benefits are not guaranteed. Incorrect, outdated or overgeneralized advice could cause crop losses, financial harm or environmental damage. High-risk recommendations involving pesticides, fertilizers or chemicals could also affect human and environmental safety. Underrepresentation of particular languages, crops, regions or farming practices may produce unequal model performance. In addition, community knowledge could be decontextualized or extracted without fair control if consent, provenance and ownership are not carefully managed. Farmers may also place excessive trust in an AI system and reduce consultation with extension practitioners or agricultural experts.

The project therefore treats fairness, dignity, autonomy, scientific integrity, cultural preservation, sustainability and community well-being as design and evaluation requirements. AgriSLM will be developed as decision support rather than an automated decision-maker. Authoritative agricultural and climate sources will be prioritized; local knowledge will be labeled according to its evidence status; high-risk advice will require stronger source grounding; and uncertain or unsupported questions should trigger clear limitations or expert escalation. Dataset provenance, consent, representation, versioning and known limitations will be documented through the project's Data Card.

Over time, AgriSLM could support more equitable access to agricultural knowledge and complement existing extension services, but it cannot solve structural constraints such as land inequality, finance, infrastructure or climate change. Its impact will therefore be evaluated through agronomic correctness, safety, regional relevance, accessibility, community usefulness and offline technical performance. Broader deployment should occur only after field validation with farmers, extension practitioners and agricultural experts demonstrates that the system provides meaningful benefit without introducing unacceptable harm.

7. Key Uncertainties and Limitations

- The exact South African pilot community, local language and priority crops remain Phase 1 validation decisions.
- Farmer requirements and perceived usefulness require direct field validation.
- Low-resource languages may have limited high-quality agricultural data.
- Official recommendations and regulatory information may change and require continuing maintenance.
- Offline deployment creates constraints on model size, update frequency and hardware performance.
- The model can improve information access but cannot independently remove wider economic, institutional or environmental barriers.

8. Impact Evaluation Indicators

Impact Area	Indicative Measure
Agronomic safety	Expert-reviewed correctness and zero tolerance target for unsupported high-risk recommendations.
Fairness	Performance reported separately across represented regions, crops and languages.
Accessibility	Farmer comprehension, language quality and successful low-connectivity/offline use.
Autonomy	Evidence that users understand AgriSLM as decision support and can identify uncertainty/escalation messages.
Community benefit	Farmer and extension-practitioner ratings of relevance, practicality, trust and usefulness.

Environmental responsibility	Quality and safety of soil, water, climate-resilience and sustainable-production guidance.
Accountability	Traceable sources, dataset versions, correction procedures and documented limitations.

9. Final Impact Principle

AgriSLM should be considered successful only if its technical performance translates into safe, accessible, locally relevant and demonstrably useful agricultural decision support for the communities it is intended to serve.

10. Basis of this Document

This impact statement was prepared from the project's established Values-Led Problem Statement, AgriSLM Research Framework, Ethical Dataset Data Card and the TRI AI Impact Statement Card task. It should be updated as Phase 1 field evidence, community feedback, dataset permissions and technical evaluation results become available.